



MACHADO BOT: GENERATING TEXTS LIKE MACHADO DE ASSIS

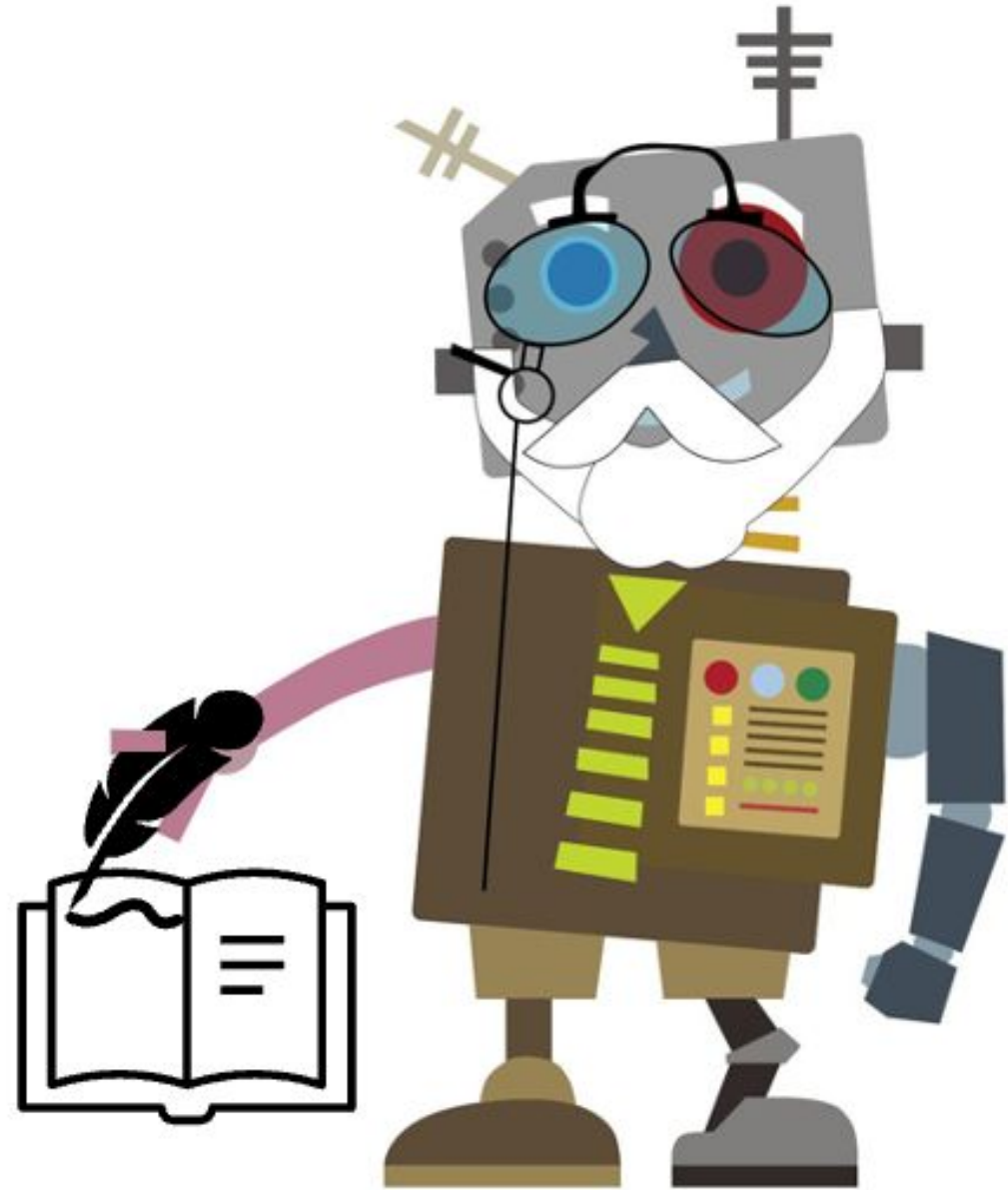
An Introduction to Language
Models
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MACHADO BOT

What is Machado Bot?

A model trained to generate text in the style of Machado de Assis. He uses texts extracted from books such as *Dom Casmurro* and *Memórias Póstumas de Braz Cubas*.

Simplified introduction to Large Language Models (LLMs) such as GPT.



<https://github.com/Mjrovai/MachadoAssisBot>

DATA PREPARATION

Data was collected from 7 books by Machado de Assis (**2.4 million characters**).

Preprocessing: Removal of irrelevant characters and structuring of the text for analysis.

Importance of clean data for training.

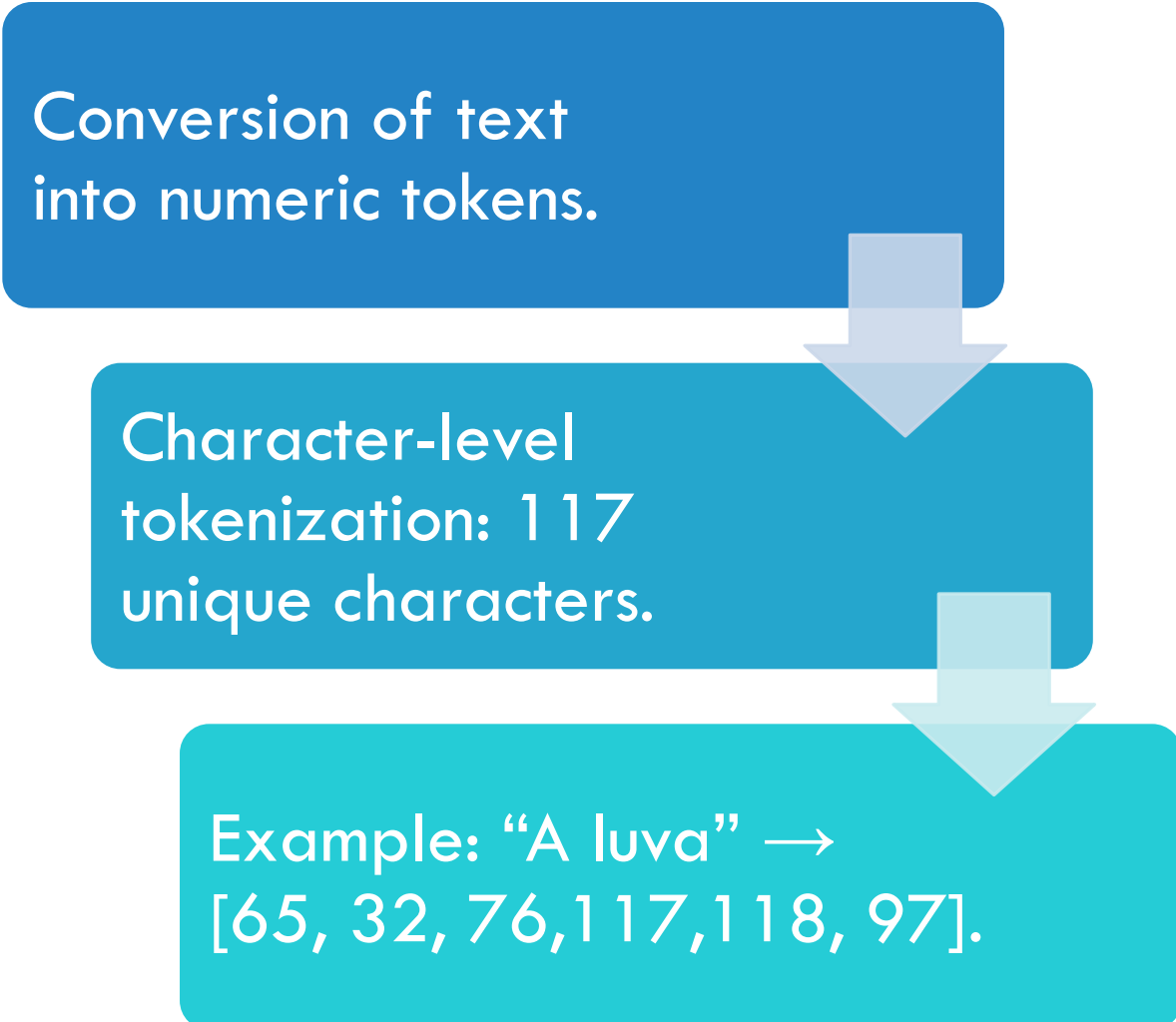


Project Gutenberg

Memorias Posthumas de Braz Cubas, Dom Casmurro, Quincas Borba, Papeis Avulsos, A Mão e a Luva, Esaú e Jacob, and Memorial de Ayres.

TOKENIZATION AND VOCABULARY

Conversion of text
into numeric tokens.



```
graph TD; A[Conversion of text into numeric tokens.] --> B[Character-level tokenization: 117 unique characters.]; B --> C[Example: "A luva" → [65, 32, 76, 117, 118, 97].];
```

Character-level
tokenization: 117
unique characters.

Example: "A luva" →
[65, 32, 76, 117, 118, 97].

TOKENIZATION

```
['\n', ' ', '!', '"', '$', '&', "'", '(', ')', '*', '+', ',', '-',  
'.', '0', '1', '2', '3', '4', '5', '6', '7', '8', '9', ':', ';',  
'=', '?', 'A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'J', 'K',  
'L', 'M', 'N', 'O', 'P', 'Q', 'R', 'S', 'T', 'U', 'V', 'W', 'X',  
'Y', 'Z', '[', ']', '_', 'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h',  
'i', 'j', 'k', 'l', 'm', 'n', 'o', 'p', 'q', 'r', 's', 't', 'u',  
'v', 'w', 'x', 'y', 'z', 'š', '«', '°', '»', 'À', 'Á', 'Ã', 'Ç',  
'É', 'Ê', 'Ë', 'Í', 'Ó', 'Ú', 'à', 'á', 'â', 'ã', 'æ', 'ç', 'è', 'é',  
'ê', 'í', 'î', 'ñ', 'ò', 'ó', 'ô', 'õ', 'ú', 'û', 'æ', '—', '']
```

```
[31, 42, 40, 1, 30, 28, 46, 40, 48, 45, 45, 42, 0, 0, 36, 0, 0,  
31, 71, 1, 76, 65, 76, 77, 68, 71, 13, 0, 0, 48, 69, 57, 1, 70,  
71, 65, 76, 61, 1, 60, 61, 75, 76, 57, 75, 11, 1, 78, 65, 70, 60,  
71, 1, 60, 57, 1, 59, 65, 60, 57, 60, 61, 1, 72, 57, 74, 57, 1,  
71, 1, 32, 70, 63, 61, 70, 64, 71, 1, 41, 71, 78, 71, 11, 1, 61,  
70, 59, 71, 70, 76, 74, 61, 65, 1, 70, 71, 0, 76, 74, 61, 69, 1,  
60, 57, 1, 30, 61, 70, 76, 74, 57, 68, 1, 77, 69, 1, 74, 57, 72,  
57, 82, 1, 57, 73, 77, 65, 1, 60, 71, 1, 58, 57, 65, 74, 74, 71,  
11, 1, 73, 77, 61, 1, 61, 77, 1, 59, 71, 70, 64, 61])
```

TRAINING SEQUENCES



Goal: Predict the next character in a sequence.



Length of the sequence: 150 characters (paragraph).



Input 'Boa tarde, meu nom'

Output 'oa tarde, meu nome'.

EMBEDDING

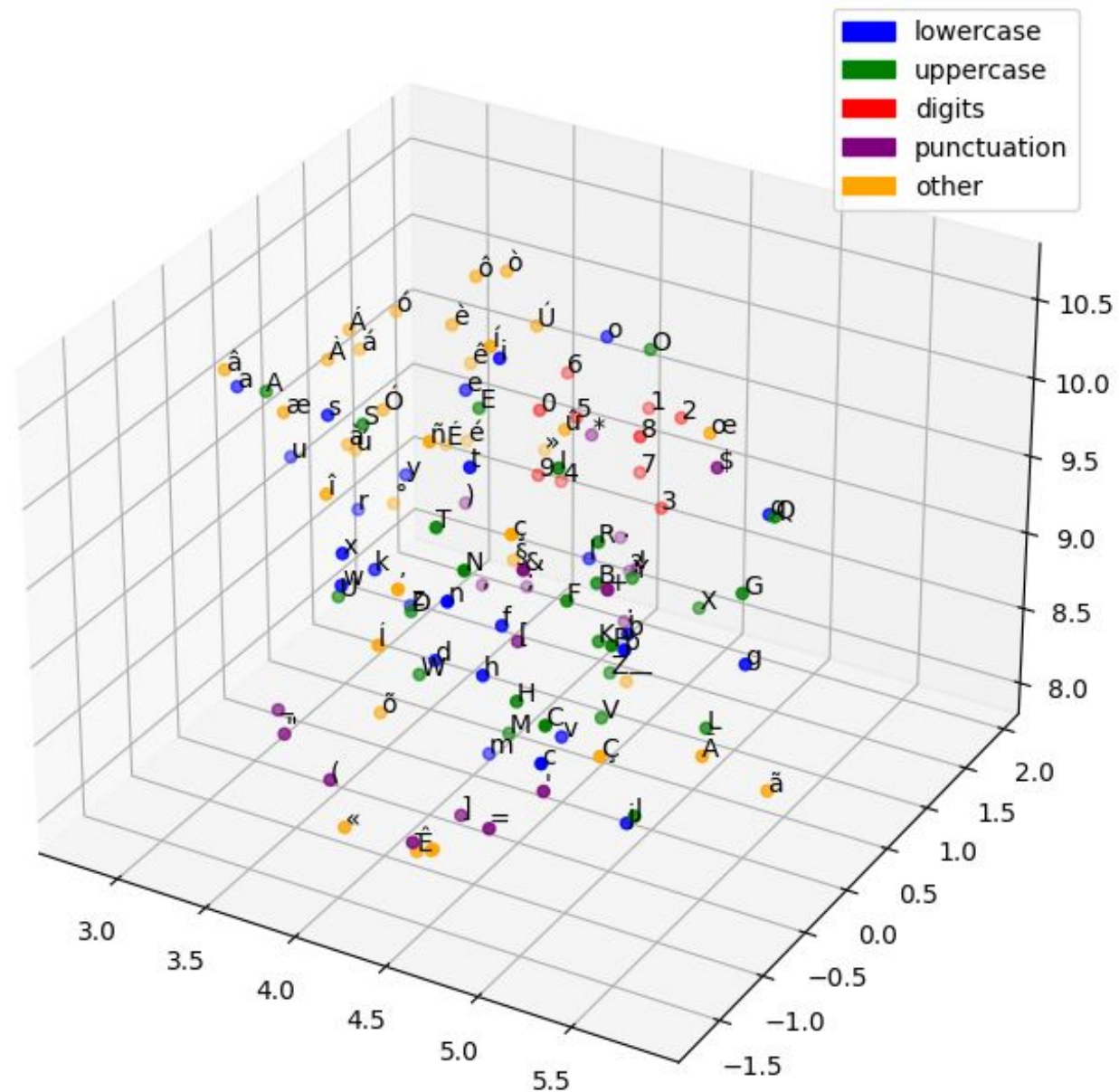
Each character is represented
as a vector of 256
dimensions.



Embedding captures
relationships between
characters in dense vectors.

EMBEDDING

3D Projection of Character Embeddings Using UMAP



Word2Vec - Embedding Projector

MODEL ARCHITECTURE

Embedding Layer:
Converts characters into
dense vectors.



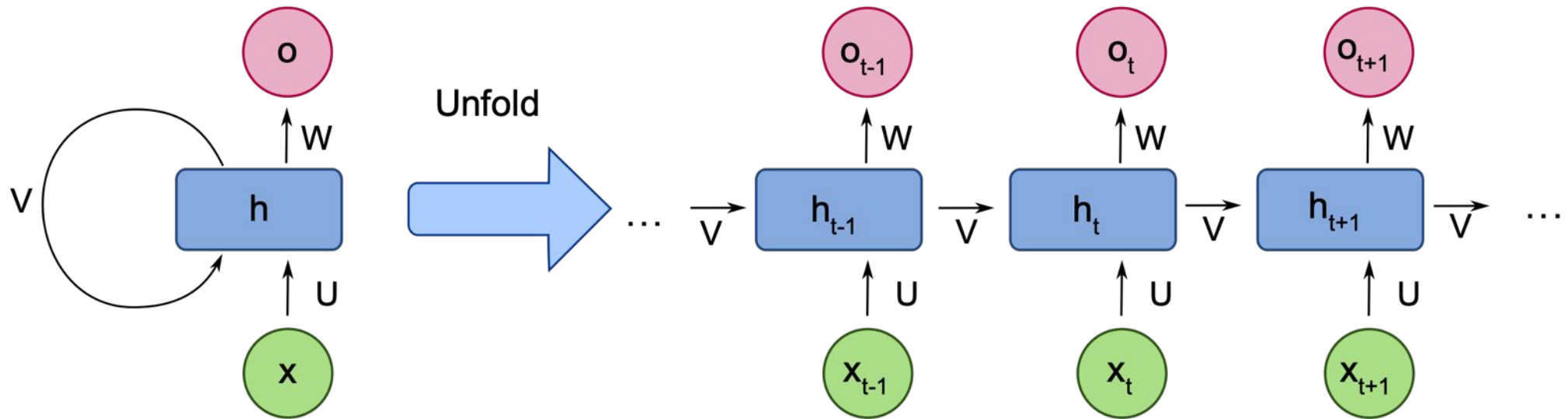
RNN/GRU Layer (1024
units): Learn from
sequences.

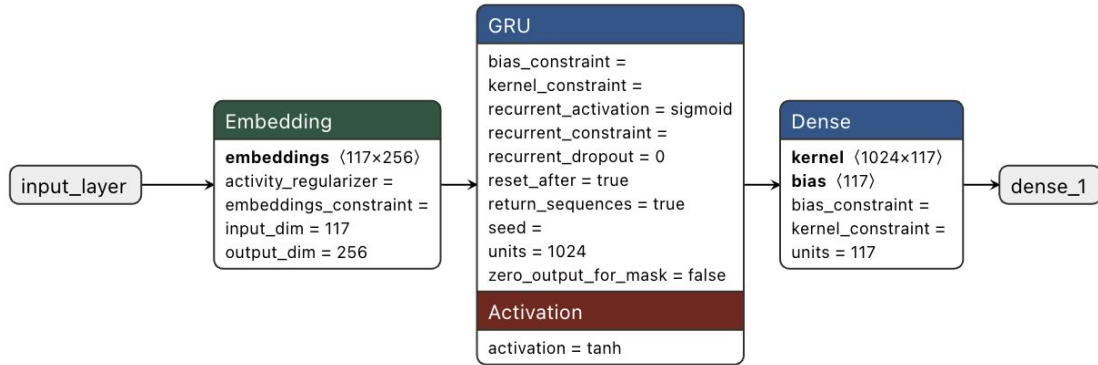


Dense layer: Generates
probabilities for each
character (117).

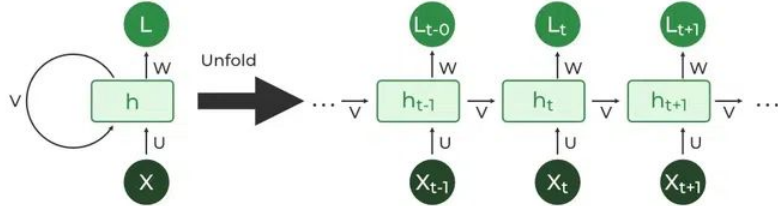
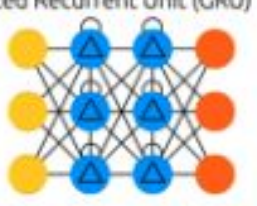
Deep Learning models (or artificial neural networks)

Recurrent Neural Networks (RNNs): Designed for **sequential data like time series or text**, these networks use their internal state (memory) to process sequences of inputs.





Gated Recurrent Unit (GRU)



Model: "sequential"

Layer (type)	Output Shape	Param #
embedding (Embedding)	(1, 150, 256)	29,952
gru (GRU)	(1, 150, 1024)	3,938,304
dense (Dense)	(1, 150, 117)	119,925

Total params: 4,088,181 (15.60 MB)
 Trainable params: 4,088,181 (15.60 MB)
 Non-trainable params: 0 (0.00 B)

RNN MODEL (RECURRENT)

MODEL TRAINING

Loss Function: Categorical
Sparse Crossentropy

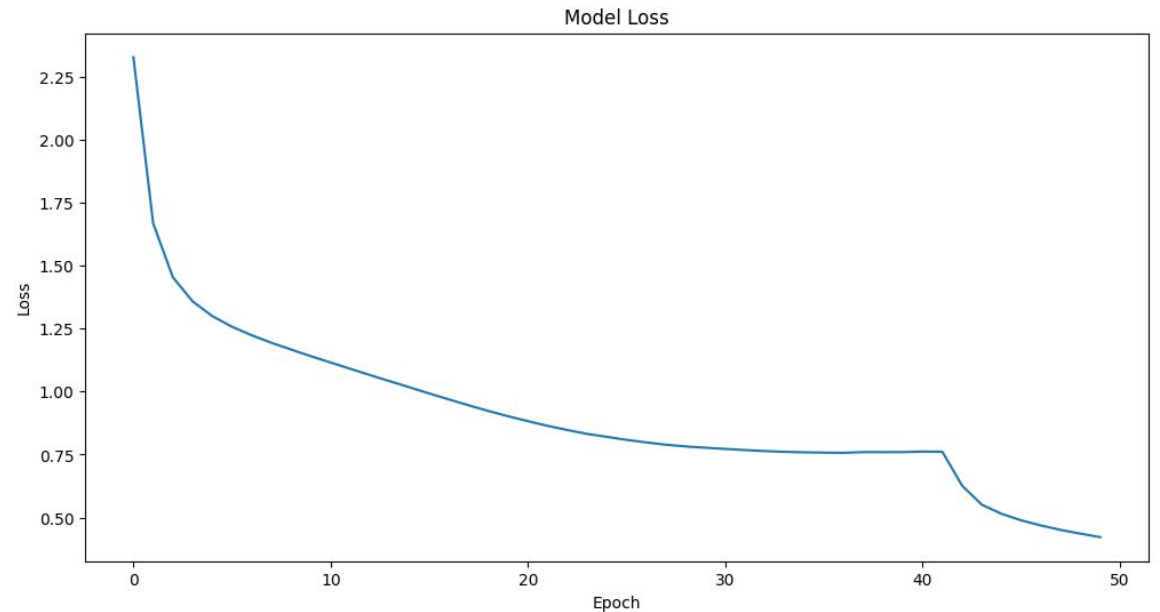
Optimizer: Adam

Epochs:50

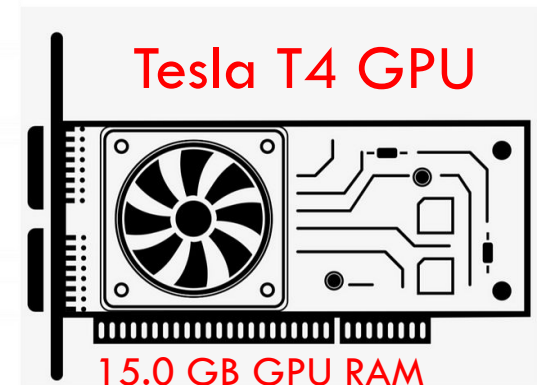
lot size: 64

buffer size: 10,000

Monitoring loss reduction over
time.



(20 to 25 minutes for training)



TEXT GENERATION

The template generates text character by character from an initial text:

"A LUVA DE CASMURRO".

Temperature controls randomness (0.5 for predictable, 1.0 for creative text).

Generated text with temperature 1.0:

A LUVA DE CASMURRO NOEFPA

E presidente da Gloria e José Dias?

Carlos Maria abaixou os olhos e entrando os seus gestos, para a escolha da madrugada, quando lhe falavam baixo e levou-a, como promettia.

--Virgem Maria hoje Refferes, ao contrario do jumento, ponderou Paulo.

--Então, panejar outro sentimento, que é tudo isso, uma vez ou o conselheiro Xavier, onde achar, patriota. Não sentira dous descansamos que a tenha separação com que elle correspondi á viuva.

--A morte é outro capitulo.

O que aquillo era do caso, que a recebeu sem nada, posto que, sendo fui distribuido muita vez o relógio foi cafa hesitou mais do que a força lhe deu outro ponto em que a deixara tão contraria. Vim com ella tambem a noticia e a hora exacta em commum, repetiram a confessara virtude, e era verdade. Para que mandasse o phenomeno,--eu, que tempo lhe pedia então, aos não deixar escripto o negocio, restringos, não saberia rugeiro.

CHALLENGES AND LIMITATIONS

*Limited **context window** (150 characters).*

Difficulty in maintaining coherence in long texts.

Character-level modeling vs. word-level modeling.

CONNECTING WITH MODERN LANGUAGE MODELS



Our Model (MachadoBot) :

- Training data: **2.4 million characters (bytes)** (7 books).
4 million parameters,
Character-level tokenization (**117**)
RNN architecture.

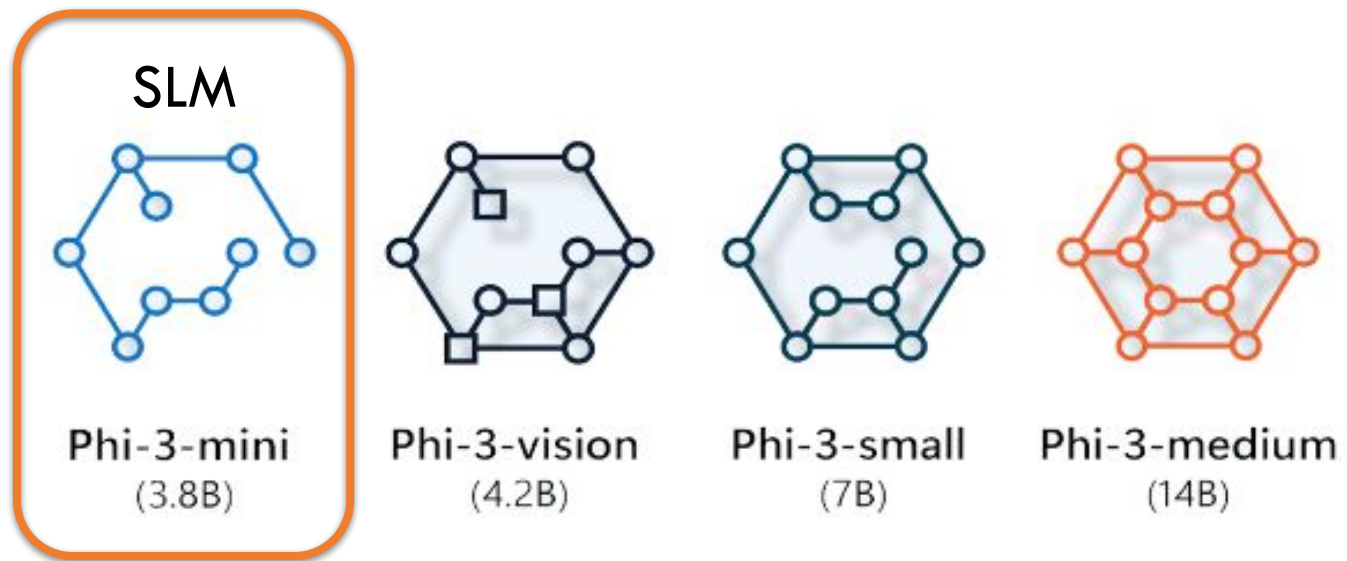


Open AI GPT-3 (2020):

- Training data: **45 Trillion bytes** (text)
175 billion parameters,
Subword tokenization (**2,048 tokens**),
Transformer Architecture.



Modern models handle long-range dependencies better.



- **Architecture: Transformer – 3.8 Billion Parameters**

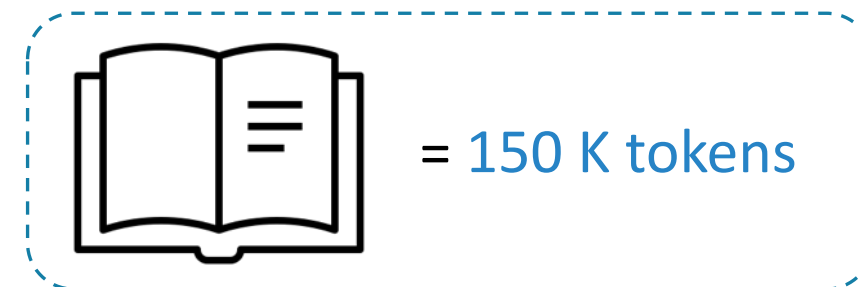
Inputs: Text.

Context length: 128k tokens

GPU: 512 H100-80G

Training time: 7 days

Training data: 3.3 Trillion tokens**



~ 350 pages



~ 300 words/page



1 word = ~ 1.4 token

**** Equivalent to 23 million books, that is:
17% of All the books in the world**

Questions?

